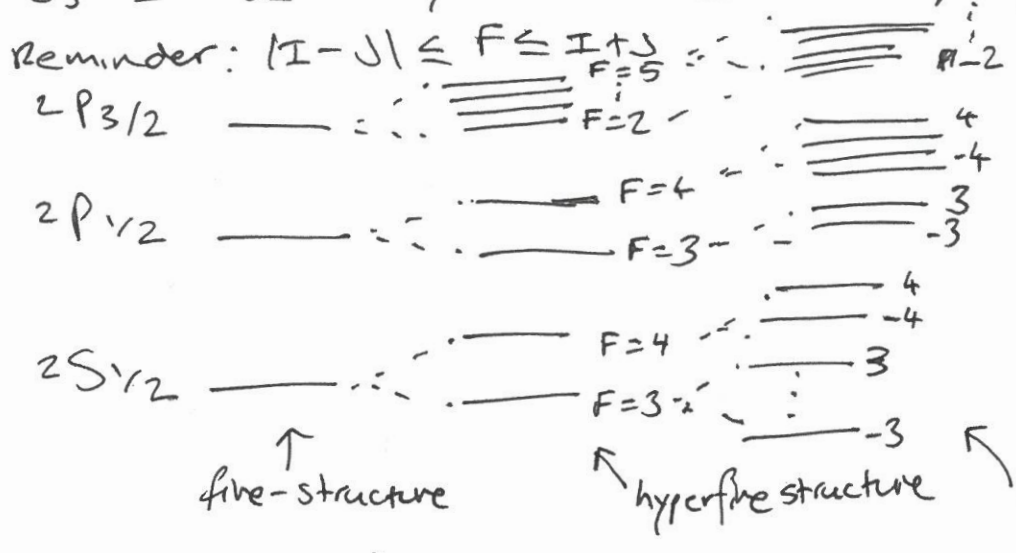
PHYS4055 tutorial week 12: 22/10.

1. MOT for Cs ground state for P_J excited states. What transition is best?

^{133}Cs $I = [redacted]\ 7/2$, HFS: $^2S_{1/2}$ $F = 3, 4$; $^2P^\circ_{1/2}$ $F = 3, 4$; $^2P^\circ_{3/2}$ $F = 2, 3, 4, 5$

Reminder: $|I - J| \leq F \leq I + J$



E1 (elec. dipole) transition

$\langle f|d|i\rangle$: $\Delta\pi = \pm 1$ (parity), $\Delta J = 0, \pm 1$ (no $0 \to 0$), $(\Delta L = \pm 1)$, [redacted], $\Delta S = 0$

(assume $q = 0$)

$\Delta m_J = 0$

[redacted]

$\Delta m_F = 0$

Possible transitions:

$S_{1/2}$ $F=3$ —excit.→ $P^\circ_{1/2}$ $F=3$ —de-excit.→ $S_{1/2}$ $F=3,4$

→ $P^\circ_{1/2}$ $F=4$ → $S_{1/2}$ $F=3,4$

$S_{1/2}$ $F=4$ → $P^\circ_{1/2}$ $F=3$ → $S_{1/2}$ $F=3,4$

→ $P^\circ_{1/2}$ $F=4$ → $S_{1/2}$ $F=3,4$ (A)

$S_{1/2}$ $F=3$ → $P^\circ_{3/2}$ $F=2$ → $S_{1/2}$ $F=3$ ← (original state)

→ $P^\circ_{3/2}$ $F=3$ → $S_{1/2}$ $F=3,4$

→ $P^\circ_{3/2}$ $F=4$ → $S_{1/2}$ $F=3,4$

$S_{1/2}$ $F=4$ → $P^\circ_{3/2}$ $F=3$ → $S_{1/2}$ $F=3,4$

→ $P^\circ_{3/2}$ $F=4$ → $S_{1/2}$ $F=3,4$ (B)

→ $P^\circ_{3/2}$ $F=5$ → $S_{1/2}$ $F=4$ ← (original state)

[redacted]

(A): $S_{1/2}$ $F=3$: [redacted] $m_F = \pm 3, \pm 2, \pm 1, 0$ → $P^\circ_{3/2}$ $F=2$ $m_F = \pm 2, \pm 1, 0$

(B) $S_{1/2}$ $F=4$: $m_F = \pm 4, \ldots, 0$ → $P^\circ_{3/2}$ $F=5$ $m_F = \pm 5, \ldots, 0$

$= 2F+1$ projections (m_F) per F

= (A) has [redacted] 'dark' states: transitions where no excitation from $S_{1/2} \to P^\circ_{3/2}$

∴ (B) is the better transition

e.g. $S_{1/2}$ $F=3$ $m_F=\pm 3$ $\xrightarrow{\sigma_+}$ $P^o_{3/2}$ $F=2$ $m_F=+2$ (not $\sigma_\pm$ allowed by $\Delta m_F = 0$)

$S_{1/2}$ $F=4$ $m_F=+4$ $\xrightarrow{\sigma_\pm,\pi}$ $P^o_{3/2}$ $F=5$ $m_F=+5,+4,+3$

Is there a contribution from $D_{3/2}$ & $D_{5/2}$?

e.g. could there be a: $S_{1/2}$ $F=3$ $\xrightarrow[\text{excit.}]{}$ $P^o_{3/2}$ $F=2$ $\xrightarrow[\text{de-excit.}]{}$ $D_{3/2}$ $F=2$?

$E(P^o_{3/2}) < E(D_{3/2})$

$\Rightarrow$ no spontaneous emission

2. Atomic beam of ^{132}Cs at 300 K slowed by laser ($\lambda = 852$ nm)
What is the stopping distance at half max. deceleration?

Recall from lectures:

= most probable velocity for a gas: $v_0 = \sqrt{\frac{2k_BT}{m}}$

= recoil velocity : $v_r = \frac{h}{m\lambda}$

= max. deceleration : $a_{max} = \frac{v_r}{2\tau}$

Note 1 amu $\approx 1.66 \times 10^{-27}$ kg

Get: $v_0 \approx 194\ ms^{-1}$, $v_r \approx 3.5 \times 10^{-3}\ ms^{-1}$, $a_{max} \approx 5.92 \times 10^4\ ms^{-2}$

Stopping distance: $z = \frac{v_0^2 - v^2}{a_{max}} \approx 0.64\ m$

($v^2 = v_0^2 - 2za$, $a = \frac{a_{max}}{2}$)

Number of photons scattered: $N = \frac{t}{\tau} = \frac{z/(v_0/2)}{\tau} \approx 2.2 \times 10^5$ photons

3. Why is it difficult to construct lasers in UV range?

lasers need population inversion (more atoms in upper than lower)

E1 transition (spontaneous emission): $A_{ab} \sim \omega^3$ (*)

UV: $\lambda \sim 10^{-8}$ m (10 nm) (vis./IR ~ 100–1000 nm)

$E = h\omega$: large ω required for transition

= UV transitions by (*) more likely to spontaneously emit

∴ loss of photons ∴ loss of population inversion.

4. a) $\tau_A < \tau_B$ (lifetime). Which can be cooled at a higher rate? Which can be cooled to a lower limit?

= Shorter lifetime leads to greater scattering => faster cooling (Atom A can be cooled faster)

= Doppler limit: $k_B T_D = \frac{\hbar\Gamma}{2}$ ∴ atom B can be cooled to a lower Doppler limit
↳ $\Gamma \propto 1/\tau$

b)

Note: $v_r = \frac{h}{(m\lambda)}$ (lower λ => greater photon energy /momentum transfer)

∴ C has larger recoil velocity

c) What makes a transition good for laser cooling?

1. low excited-state lifetime => large scattering rate
2. closed transition / loop or ≈ 2 ~~a~~ level system
3. Reasonable transition frequency